

An introduction to measure theory - T.Tao

Statistical Learning Theory Group

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Preface

Excercise 0.0.1. *if $(x_\alpha)_\alpha$*

Proof. It is an easy proof.

□

Excercise 0.0.2. *Exercise 0.0.2*

Chapter 1

Measure Theory

1.1 Prologue: The problem of measure

Excercise 1.1.1. *Exercise 1.1.1*

Lemma 1.1.2 Let $E \subset \mathbb{R}^d$ be an elementary set. (ii) If $E = \cup_{i=1}^n B_i$, and $E = \cup_{j=1}^m C_j$ Then $\sum_{i=1}^n |B_i| = \sum_{j=1}^m |C_j|$

Proof

First, we need prove $B_i \cap C_{j,i,j}$ are boxes. Considering

$$\begin{aligned} B_i &= I_{i,1} \times \cdots \times I_{i,d} \\ C_j &= J_{j,1} \times \cdots \times J_{j,d} \\ I_{i,r} \cap J_{j,r} &= I' || \emptyset \end{aligned}$$

which implies $B_i \cap C_j = \text{box} || \emptyset$

Suppose overlap

$$\begin{aligned} (B_i \cap C_j) \cap (B'_i \cap C'_j) &\neq \emptyset \\ \implies x \in B_i \cap B'_i \text{ and } x \in C_j \cap C'_j \end{aligned}$$

But B_i, B'_i disjoint which cause a contradiction. Hence, $B_i \cap C_j$ mutually disjoint.

$\forall x \in E$, always $\exists i$ that $x \in B_i$ and j that $x \in C_j$ This means $x \in B_i \cap C_j \implies E \subseteq \cup_{i,j} (B_i \cap C_j)$

We also have

$$B_i \cap C_j \subseteq E \implies \cup_{i,j} (B_i \cap C_j) \subseteq E$$

Combine these equations , we have

$$E = \sqcup_{i,j} (B_i \cap C_j)$$

Apply measure function and sum up with double indexing result the desired equality.